

Arteriovenous malformations: Evaluation and treatment

BYUNG-BOONG LEE, JAMES LAREDO, RICHARD F. NEVILLE, AND ANTON N. SIDAWY

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56.1 INTRODUCTION

Arteriovenous malformation (AVM) is a congenital vascular malformation (CVM) in which an anatomic defect results in shunting of arterial blood to the venous system to varying degrees.^{1–3} The arteriovenous (AV) communications allow the shunting of high-velocity, low-resistance flow from the arterial vasculature into the venous system.^{4–6}

This unique condition of “AV shunting” between the arterial and venous systems causes altered cardiovascular hemodynamics centrally, peripherally, and locally, thus making the AVM the most hemodynamically complex type of CVM.^{7–9}

The hemodynamic alterations to the arterial, venous, and lymphatic systems affect the entire cardiovascular system and may produce cardiac failure, peripheral arterial insufficiency (e.g., gangrene), chronic venous insufficiency, and lymphatic overload due to venous hypertension. These lesions also have local effects on the surrounding tissues and organs and are associated with significant morbidity and high recurrence rates following treatment.^{4,5} The AVM is considered to be the most virulent of all CVMs.^{10,11}

56.2 INCIDENCE AND EPIDEMIOLOGY

Epidemiologic data reported in the medical literature often misrepresent the true incidence and prevalence of the CVMs based on name-based classifications. Before the International Society for the Study of Vascular Anomalies (ISSVA) and Hamburg CVM Classification systems were

established, much confusion remained with limited reliability of data regarding AVM incidence and prevalence.^{5,12,13}

Data reported by Tasnadi (1993) show the overall incidence of CVMs to be 1.2%.¹⁴ The CVMs are a group of various birth defects involving the vascular system that present at birth over 90% of the time, with a male–female ratio of 1:1.^{1,4,12}

The incidence and prevalence of cerebrospinal AVMs were reported in a systematic review.¹⁵ The incidence of AVMs was found to be approximately 1 per 100,000 per year in unselected populations, and the point prevalence in adults was approximately 18 per 100,000. In the United States, 1 per 250,000 people has a cerebrospinal AVMs, appearing in the age range of 20–40 years, with a male–female ratio of 1:1 or 1:2.¹⁵

Peripheral AVMs were reported to be the least common CVMs, representing approximately 10%–15% (range of 5%–10% to 15%–20%) of all clinically significant CVM lesions.^{4,16}

Among all CVMs as defined by the Hamburg Classification (Table 56.1),^{5,13} the AVM is a much more rare condition than venous malformations (VMs)^{17,18} or lymphatic malformations (LMs).^{19,20} The majority of CVMs are either VMs or LMs, and the VMs represent approximately two-thirds of all CVMs.²¹

Among the AVMs, the “extratruncular” AVM (formerly angiomatous AVM) lesions comprise most AVM cases.⁴ The “truncular” AVM lesions are extremely rare and occur as a result of direct communication between an artery and a vein; for example, between the pelvic vessels or the femoral

Table 56.1 Hamburg Classification of congenital vascular malformations**Primary classification^a**

- Arterial malformation
- Venous malformation
- Arteriovenous malformation
- Lymphatic malformation
- Capillary/microvascular malformation
- Combined vascular malformation: hemolymphatic malformation

Embryological sub-classification^bExtratruncular forms (former “angioma”)^c

- Diffuse, infiltrating
- Limited, localized

Truncular forms^c

- Obstruction and/or stenosis
 - Aplasia, hypoplasia, hyperplasia
 - Membrane, congenital spur
- Dilatation
 - Localized (aneurysm)
 - Diffuse (ectasia)

^a Modified based on the consensus on congenital vascular malformations through the international workshop in Hamburg, Germany, in 1988 and Seoul, Korea, in 1995.

^b Developmental arrest at the different stages of embryonal life: earlier stage—extratruncular form; latter stage—truncular form.

^c Both forms may exist together, may be combined with various other malformations (e.g., capillary, arterial, arteriovenous shunting, venous, hemolymphatic and/or lymphatic), and/or may exist with hemangioma.

artery and vein. They are genuine fistulous lesions without a nidus such as a patent ductus arteriosus or pulmonary AV fistula.⁴

The vast majority of AVMs exist alone as independent lesions, but infrequently occur with a VM and/or LM, making its diagnosis and management more difficult. These mixed CVMs, newly classified as hemolymphatic malformations, often become a clinician’s nightmare (e.g., Parkes–Weber syndrome [PWS]). Their management is quite confusing and the treatment results are often disappointing (e.g., microshunting AVM).^{22,23} To date, no racial, demographic, or environmental risk factors for AVMs have been identified.⁴

56.3 ETIOLOGY AND PATHOPHYSIOLOGY

There has been significant progress to explain the growth tendency of extratruncular AVM lesions based on gene mutations occurring in tissues. Several gene mutations have been shown to be responsible for the defective development during embryogenesis resulting in high-flow AV shunts through dysplastic vessels.^{4,6,24–27}

Although further studies are required in order to better understand these mechanisms, identification of the causative genes in several defects has already allowed a more

precise diagnosis. Some mutations have been found to be responsible for a number of rare inheritable vascular malformations, such as Osler–Weber–Rendu syndrome (hereditary hemorrhagic telangiectasia), blue rubber bleb nevus syndrome (Bean’s syndrome), RASA1 mutations, PTEN mutations, etc.^{24–27} For example, hereditary hemorrhagic telangiectasia is an autosomal dominant condition caused by loss-of-function mutations in the genes encoding activin receptor-like kinase-1 (ACVRL1) and endoglin (ENG), both transforming growth factor- β vascular growth factors.

The majority of AVM lesions develop progressively over time, particularly in the brain, lungs, and liver, and are eventually diagnosed in patients when they are adults. A high percentage of the angio-architecture of the AVM is an AV fistula, and positive results have been reported in patients treated with angiogenesis inhibitors, such as bevacizumab. RASA1 mutations are responsible for another autosomal dominant condition of combined capillary malformations (CMs) and AVMs. The most common sites for AVMs among affected family members are the brain, spine, face, and extremities. Intracranial AVMs can be typical arteriovenular malformations or pial AV or arteriovenous fistulae.

PWS is associated with the RASA1 mutation. The AVMs in these patients generally present as “micro-shunting” lesions initially, remaining as diffuse small vessel lesions in the muscles and subcutaneous tissues of the limbs. Together with other vascular malformation components (VMs and LMs), it often results in marked tissue overgrowth of the affected muscle, bone, and subcutaneous fat. However, individuals with PWS without multiple additional CM lesions generally do not carry RASA1 mutations.

PTEN mutations are responsible for the focal tissue overgrowth in hamartomas that frequently contain AVMs. These mutations represent syndromes of tissue overgrowth (e.g., Cowden and Bannayan–Riley–Ruvalcaba syndromes). Multiple AVM lesions are common, often affecting the muscles of the limbs, paraspinal muscles, and the dura mater of the brain. The angio-architecture is usually arteriovenous and exhibits aggressive behavior that is difficult to control by embolization.^{4,6}

Further research on the genetic mechanisms of CVM/AVM formation is required for better understanding of the morphogenesis and biology of both extratruncular and truncular AVM lesions.

56.3.1 Abnormal endothelial cell turnover rate among the AVMs

The endothelial cell turnover rate (ECTR) is an endothelial cell characteristic that differentiates vascular malformations from hemangiomas, with the latter having higher rates. The ECTR has been found to be significantly greater in AVMs than in normal blood vessels. The mean Ki-67 index is higher for AVM vessels than control vessels, with an approximately seven-fold increase in the number of non-resting endothelial cells.^{4,6} The Ki-67 protein presents during all active phases of the cell cycle except resting phase

(G0). Therefore, non-resting endothelial cells can be identified by use of immunohistochemistry for the Ki-67 antigen.

Increased expression of stromal cell-derived factor-1 (SDF-1) was also found in the AVM nidus that suggests that endothelial progenitor cells (EPCs) may play a role in maintaining active vascular remodeling within the AVM nidus, whereas SDF-1 expression is rarely identified in normal vessels.^{4,6,28,29}

In addition, mRNA expression of the factors that recruit the EPCs vascular endothelial growth factor, SDF-1 α , hepatocyte growth factor, and hypoxia-inducible factor-1 have been found to be different in Schobinger stages II and III of the AVMs. Increased expression of EPCs and growth factors among higher-staged AVMs suggests EPCs as a promoting factor for the evolution of AVMs, stimulating their recruitment for neovascularization.^{28,29}

The research on the ECTR over the last 10 years has brought substantial improvement to our molecular vision of AVMs, together with the role of EPCs.^{4,6}

56.3.2 The different genetic expression of EPC activity

Despite the early work of Mulliken et al. indicating that vascular malformations do not grow by cellular proliferation, more recent data demonstrate the importance of cellular proliferation and the upregulation of matrix metalloproteinases, especially in symptomatic AVMs.^{4,30} This new finding may lead to the development of new strategies of pharmacotherapy for treating symptomatic AVMs.^{4,30}

56.3.3 The sporadic and familial/syndrome-based AVMs^{4,6,24–27}

The “sporadic” type of AVMs without documented genetic mutations may be explained by underlying genetic abnormalities in the future. Nevertheless, the familial/syndrome-based AVMs behave differently from the majority of AVMs belonging to the sporadic type. AVMs in patients with RASA1 mutation generally remain relatively stable with minimal progression, although symptomatology varies based on the anatomical location of the AVM lesion, as those of the central nervous system can produce significant mass effect and hemorrhage.

The AVM lesion among PWS patients often shows a slow progression when compared to the sporadic type of AVM occurring in the lower limb. The amputation rate is more frequent with sporadic AVMs in the lower limb, even though they usually present with a smaller size lesions.

Bleeding of visceral AVM lesions in patients with hereditary hemorrhagic telangiectasia (Osler–Weber–Rendu syndrome) is more frequent than in the sporadic type of visceral AVM. However, the need for intestinal resection to control bleeding is exceedingly rare. On the other hand, surgical resection to control intestinal bleeding has been reported to be higher in patients with sporadic-type AVM.

The AVMs caused by PTEN mutations are known to be the most aggressive type of AVMs, not only recurring rapidly following the embolization or resection, but also with a tendency to develop new lesions at different sites.

56.3.4 Hemodynamic consequences of AV shunting

AVMs are a unique, complex vascular conditions bypassing the normal capillary system and resulting in a “short circuit” of the normal circulation between the arterial and venous–lymphatic systems. The pivotal function of the capillary system is no longer present to maintain the delicate balance between the two hemodynamically different systems.^{1–4,6,31,32} When the capillary system is no longer present, the balance between these three uniquely different hemodynamic components of the circulatory system—arteries, veins, and lymphatics—is profoundly disrupted.

This abnormal connection between the high- and low-pressure systems forces these two circulatory systems to respond in a compensatory manner to minimize hemodynamic effects. Depending on the location and/or degree (e.g., size and flow) of the fistulous connection, the arterial and venous systems produce responses that occur in two distinct phases: the compensation period and the decompensation period.^{4,6}

During the compensation period, increased heart pump function can assist the failing arterial system by low peripheral resistance through the AV fistula and so maintain arterial flow as before, preventing peripheral tissue ischemia. However, increased heart pumping results in increased load on the venous as well as lymphatic system, and subsequently on the heart itself. Initially, the normal venous system is able to respond to this increased pressure and volume produced by the AV fistula together with the lymphatic system. However, the lymphatic system has limited capacity to assist the failing venous system due to its unique lymphodynamic mechanism based on autoregulated peristaltic circulation. Once these compensatory mechanisms involving the arterial, venous, and lymphatic systems and the heart reach their maximum capacity, the decompensation period begins.

The arterial system can no longer maintain adequate arterial blood flow to the peripheral tissues distal to the AV fistula, resulting in tissue ischemia. The venous system distal to the AVM lesion is also no longer able to maintain normal valvular function and allows retrograde blood flow. Continuous reflux further impedes normal antegrade venous flow from the peripheral tissues, resulting in severe venous hypertension and chronic venous insufficiency.

Hence, the hemodynamics of the AVM affect every vascular component of AV communication, resulting in local, peripheral, and central effects. Proper management of the various pathophysiologic effects of the AVM on the entire vascular system requires precise information regarding the close relationship between three pairs of proximal, distal, and collateral arteries and veins at the different stages of the condition.^{1–4,6}

56.4 CLASSIFICATION

A consensus workshop held in Hamburg in 1988 attempted to develop a logical classification system of CVMs, as the old, traditional nosology-based classification (e.g., Klippel-Trenaunay syndrome) failed to provide appropriate differentiation of anatomic, pathophysiologic, and clinical presentations among the various CVMs.^{4,5,13,33,34}

The group first described the morphological differences between lesions involving the main vessel trunks, often with a direct communication (“truncular” form) and lesions occurring peripherally as separate defects (AV angiomas).^{4,5} In order to avoid the confusing term of “angioma” (c.f. hemangioma), Belov et al. reintroduced an old embryologic term—“extratruncular”—for these “angiomatous” AVM lesions based on their distinctively different morphology from “truncular defects.”^{4,5,13,33,34}

The morphological difference between these two groups was further categorized according to its embryological mechanism by Belov et al. as an outcome of the developmental arrest of the vascular system during two different stages of angiogenesis. Defective development occurring in the “earlier” stage would produce primitive vascular structures in the form of a “reticular network,” while those defects occurring in the “later” stage of vascular trunk formation would produce normal, mature vascular structures.^{5,34–36}

Subsequent CVM classifications as the basis of the contemporary approach to precise evaluation, diagnosis, and therapeutic implementation were refined and later became the “modified” Hamburg Classification (Table 56.1).^{4,5,13,34}

Based on the Hamburg Consensus, Mulliken et al. also introduced another new classification for vascular anomalies to differentiate the hemangiomas as vascular tumors from the vascular malformations, and further classified vascular malformations using flow characteristics: fast-flow and slow-flow lesions. This new classification was later adopted as the ISSVA Classification (Table 56.2).^{4,5,12}

In addition, two other classifications were adopted for the AVM, namely the Schobinger Classification and the Arteriographic Classification, to improve AVM management.^{4,6,37,38}

The Schobinger Classification of AVMs (Table 56.3) is important for defining the level of clinical progression in AVM diagnosis and treatment. It provides a more accurate assessment of the AVM lesions at different clinical stages and serves as a practical guideline in their management based on the patient’s clinical status.

The Arteriographic Classification of AVMs (Table 56.4) is utilized exclusively to classify the “extratruncular” AVM lesions located in the torso and extremities based on the arteriographic morphology of the “nidus.” These radiological findings of the “nidus” represent the primitive reticular networks of dysplastic minute vessels which failed to mature into normal “capillary” vessels.^{4,6,38}

The nidus lesions are classified into three types: type I (AV fistulae), type II (arteriovenous fistulae), and type IIIa and IIIb (arteriovenulous fistulae). This classification is

Table 56.2 International Society for the Study of Vascular Anomalies (ISSVA) classification of congenital vascular malformations and vascular tumors

Vascular malformations

- Fast-flow lesions:
 - Arterial malformation
 - Arteriovenous malformation
 - Arteriovenous fistula
- Slow-flow lesions:
 - Capillary malformation (port wine stain, telangiectasia, angiokeratoma)
 - Venous malformation
 - Lymphatic malformation
 - Combined vascular malformation (CVM, CLM, CLVM, CAVM, CLAVM)

Vascular tumors:

- Infantile hemangioma
- Congenital hemangioma
- Other

Note: CVM: capillary malformation + venous malformation; CLM: capillary malformation + lymphatic malformation; CLVM: capillary malformation + lymphatic malformation + venous malformation; CAVM: capillary malformation + arteriovenous malformation; CLAVM: capillary malformation + lymphatic malformation + arteriovenous malformation.

Table 56.3 Schobinger classification of arteriovenous malformations

Stage I—quiescence: Pink–bluish stain, warmth, and arteriovenous shunting are revealed by Doppler scanning. The arteriovenous malformation mimics a capillary malformation or involuting hemangioma.

Stage II—expansion: Stage I plus enlargement, pulsations, thrill, bruit, and tortuous/tense veins.

Stage III—destruction: Stage II plus dystrophic skin changes, ulceration, bleeding, and tissue necrosis. Bony lytic lesions may occur.

Stage IV—decompensation: Stage III plus congestive cardiac failure with increased cardiac output and left ventricle hypertrophy.

helpful not only for the better management of extratruncular AVM lesions, but also for predicting outcomes of endovascular treatment.

56.4.1 Clinical implementation of the embryological concept^{4,5,13,34}

AVMs are further classified into two subtypes with different embryological characteristics—extratruncular and truncular lesions—based on the embryological stage at which the developmental arrest occurred (Table 56.1).

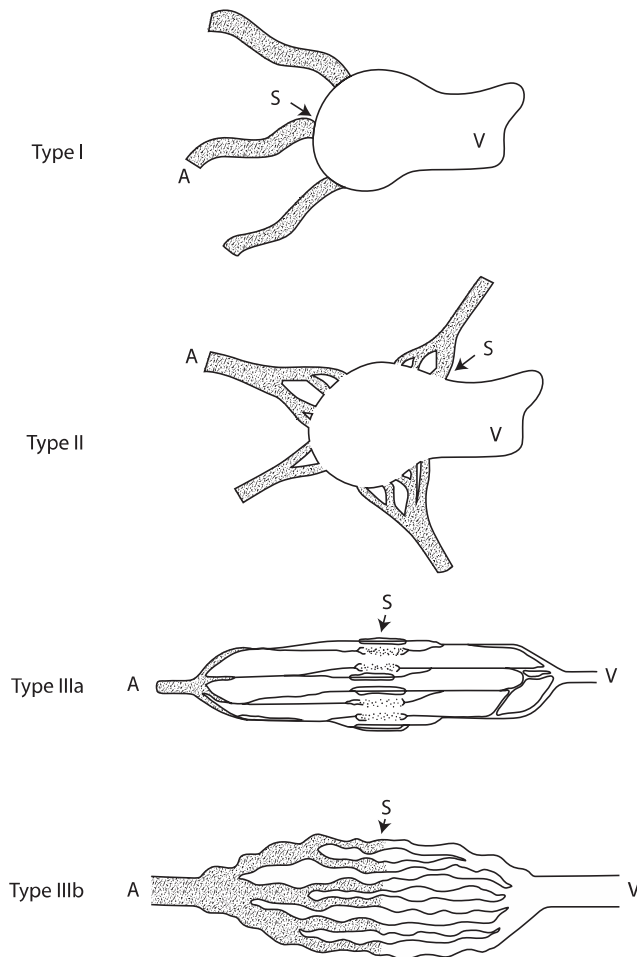
An extratruncular lesion is an embryonal tissue remnant of the primitive capillary network that is the result

Table 56.4 Arteriographic Classification of arteriovenous malformations

Type I (arteriovenous fistulae): At most three separate arteries shunted to a single draining vein

Type II (arteriovenous fistulae): Multiple arterioles shunted into a single draining vein

Type IIIa and IIIb (arteriovenous fistulae): Multiple shunts between the arterioles and venules



of developmental arrest occurring during an “early stage” of embryogenesis. This type of the endothelial cell/lesion therefore retains the characteristics of mesenchymal cells, maintaining its ability and potential to proliferate, making its clinical and biological behavior unpredictable.^{35,36}

Stimulation of an extratruncular lesion by various intrinsic (e.g., menarche or pregnancy) and extrinsic (e.g., trauma or surgery) stimuli results in lesion recurrence or progression. The potential for further proliferation underscores the importance of a well-planned treatment regimen.

Stimulation of a dormant extratruncular lesion during treatment often leads to an erratic response with explosive growth, worsening the clinical picture. An infiltrating extratruncular lesion is far more complicated than a truncular lesion because of its embryologic characteristics. Therefore, this extratruncular lesion carries a higher risk of progression and has more destructive potential.

In contrast, a truncular lesion, which is the result of developmental arrest occurring during a “late stage” of embryogenesis, no longer has the potential to proliferate (e.g., pulmonary AV fistula). However, a truncular lesion is hemodynamically more significant than its extratruncular counterpart, owing to its unique high-flow “fistulous” state with no nidus in the capillary beds to limit blood flow. These types of fistulous lesions have a direct connection between the arterial and venous systems, resulting in significant hemodynamic effects centrally, peripherally, and locally (briefly, on the entire cardiovascular system) with varying degrees of severity.

56.4.2 AV fistula versus AVM^{4,5,12}

Due to the sub-classification of the AVM into AV fistula and AVM by ISSVA Classification 12, many mistakenly identify AVMs as “non-fistulous” lesions (Table 56.2). All AVM lesions are “fistulous” whether they have a “nidus” or not. In other words, there is no such AVM lesion *without* a fistulous connection between the artery and vein to allow free “high-flow” shunting. Due to the unique embryological nature of AVMs as a consequence of defective development, there are no “non-fistulous” AVMs. All AVMs are “fistulous” by nature.

The “AV fistula” lesion defined by the ISSVA Classification is equivalent to the “truncular” AVM lesion with no nidus (e.g., bicus aortae or pulmonary AVM) defined by the Hamburg Classification. The “AVM” defined by the ISSVA Classification is equivalent to the “extratruncular” AVM lesion with nidus defined by the Hamburg Classification.

56.4.3 “Nidus” versus “non-nidus” AVM^{4–6}

The term of “nidus” is a clinical radiological term used to describe the clusters of small-sized AV connections—fistulae—filled with contrast on arteriography (Table 56.4).^{4–6} “Nidus” is therefore not a histological term, nor an anatomic or pathologic term, but a descriptive term of a conglomerate of blood vessels constituting the AVM.

A “nidus” is a characteristic of an “extratruncular” AVM lesion, always being present in an extratruncular lesion, and only present in “extratruncular” AVM lesions. A “nidus” often appears as a net of pulsating, dysplastic, tortuous vessels between the artery and vein. The “nidus” remains a “diffuse, multiple small fistulous” entity, in contrast to the truncular lesion, which is a large, individual AV fistula. In contrast, truncular AVM lesions have a direct connection between the artery and vein.

56.5 EVALUATION

56.5.1 General overview

Progression of an AVM invariably produces symptoms (pain, ulceration, and bleeding) and tissue ischemia when increased shunting results in arterial steal and venous hypertension, both reducing tissue perfusion.^{1–4}

The clinical manifestations of AVMs are dependent on their anatomical location: cardiac failure for centrally located lesions and arterial and venous insufficiency associated with local venous hypertension in peripherally located lesions. In addition, local effects of AVMs may include ulceration and gangrene.^{1–4} Therefore, a complete systematic evaluation is required after a thorough history and physical examination, followed by noninvasive diagnostic imaging in order to distinguish AVMs from other CVMs.

Although most AVMs occur as single lesions, the evaluation of an AVM should be started as an evaluation of a CVM, followed by a more specific evaluation and confirmation of an AVM. The evaluation should follow basic differential diagnoses among the various CVMs.

The investigations should rule out the co-presence of other CVMs as well, since AVMs may exist combined with other CVMs, like VMs and LMs (e.g., PWS).^{1–4}

Appropriate differential diagnosis, therefore, should be made initially with various combinations of noninvasive to less invasive tests and more specific diagnostic procedures, followed by further precise and detailed assessment of the AVM as a whole (Table 56.5).^{4,6}

In addition to the assessment of the primary AVM lesion, evaluation of its secondary impact on the non-vascular organ systems, especially the musculoskeletal system, is also warranted. Early detection of vascular–bone syndrome with long-bone length discrepancy is essential for appropriate management.^{39,40}

Among the noninvasive tests, duplex ultrasound sonography (DUS) remains the first choice in the initial clinical assessment and subsequent follow-up. DUS readily differentiates the AVM, with its unique finding of pulsatile flow, from VMs and LMs. In addition, B-mode and Doppler findings of multiple vascular channels with a honeycomb appearance are characteristic of an AVM and are absent in vascular tumors.^{4,6,41}

Magnetic resonance imaging (MRI) remains the major diagnostic study for the entire group of CVMs, including the AVMs, since MRI is able to provide basic information

on lesion extent, severity, and anatomic relationship with the surrounding tissues, structures, and organs. However, standard MRI is usually not able to precisely demonstrate the nidus and is limited in terms of demonstrating the presence of dilated fast-flow vascular channels.^{4,6,42}

Transarterial lung perfusion scintigraphy (TLPS)⁵ has a unique role in determining the degree of shunting through an extremity AVM. It can detect and assess a micro-AV shunting lesion. These types of malformations are notoriously difficult to detect with conventional arteriography alone. Such micro-AVM lesions are often missed with conventional arteriography and frequently occur in the combined form of CVM (e.g., PWS).^{2,6,8}

In addition, TLPS can also provide quantitative measurements of the shunting status during therapy; TLPS may replace the substantial role of traditional arteriography as a follow-up assessment tool for extremity AVMs.^{2,6,8}

56.6 TREATMENT STRATEGY

56.6.1 General principle

Treatment of AVMs, either extratruncular or truncular, should take priority over all other CVMs because of the potential for the development of life- and/or limb-threatening conditions.^{1,4,32}

The main goal of the treatment of all “extratruncular” AVM lesions should be to eliminate the “nidus.” Incomplete surgical resection or simple occlusion and/or ligation of feeding arteries, leaving the nidus intact, is inadequate treatment, resulting in stimulation of the lesion and causing its recurrence from the development of new feeding vessels.

Early aggressive control of the nidus is required to prevent recurrence and produce eventual deterioration and eradication of the AVM lesion. This is often difficult due to the high likelihood of recurrence following a conventional approach with currently available treatment modalities.

A “controlled” aggressive approach should be exercised in the treatment of AVMs only when the benefit exceeds the associated morbidity of the proposed treatment.^{4,32} The temptation to initially intervene radically in an AVM must be tempered with a realistic assessment of the long-term goal of the treatment plan.

To achieve this goal, the treatment strategy should be based on a consensus reached by the many different specialists involved in the patient's care (e.g., vascular surgeon, orthopedic surgeon, plastic surgeon, head and neck surgeon, interventional radiologist, and physiatrist). The final decision for treatment, as well as selection of the treatment modalities, should be made via a multidisciplinary team approach based on the various indications outlined above.^{2,4,7,32}

The new approach to the treatment of AVMs as a multidisciplinary team has resulted in significant improvements in diagnosis and treatment, with substantial reductions in morbidity, mortality, and recurrence rates through the full integration of the latest treatment modalities.^{7,10,32}

Table 56.5 Lists of diagnostic tests for arteriovenous malformations

Noninvasive to minimally invasive tests for initial studies:

- Duplex ultrasonography (arterial and venous)²⁵
- Whole body blood pool scintigraphy²⁶
- Transarterial lung perfusion scintigraphy⁵
- Magnetic resonance imaging of T1 and T2 images and magnetic resonance angiography²⁷
- Computed tomography (CT) and CT angiography with contrast enhancement, and/or three-dimensional CT²⁸

Invasive tests for the confirmation of the final diagnosis^a:

- Selective and super-selective arteriography
- Percutaneous direct puncture arteriography
- Standard direct puncture phlebography

^a Confirmation of the final diagnosis should be made to create a road map for the proper treatment.

Table 56.6 Indications for the treatment of arteriovenous malformations

- Hemorrhage
- High-output heart failure
- Secondary arterial ischemic complications
- Secondary complications of chronic venous hypertension
- Lesions located at a life-threatening region (e.g., proximity to the airway) or located in an area threatening vital functions (e.g., seeing, eating, hearing, or breathing)
- Disabling pain
- Functional impairment
- Cosmetically severe deformity
- Vascular–bone syndrome: abnormal long-bone growth with leg length discrepancy^{19,20}
- Lesions located in an area with potentially high risk of complication (e.g., hemarthrosis)

For advanced diagnosis and treatment of AVMs, a fully integrated specialty team is required in order to provide maximum coordination among the various CVM-related specialists, utilizing the full spectrum of endovascular treatments in addition to traditional surgical treatment.

After assessment of the extent and severity of the AVM has been performed, indications for treatment are determined based on its urgency. These indication criteria were formulated based on the “extratruncular” lesions, since the absolute majority are of the extratruncular type, while the truncular AVM lesions are extremely rare (Table 56.6).^{2,4}

Among the criteria, the latter five indications are common conditions that clinicians will encounter, but the first five indications are relatively rare, often reflecting conditions with a high risk of morbidity as well as mortality.

The role of careful assessment and diagnosis in the development of a treatment strategy that maximizes the risk–benefit ratio is critical and cannot be overemphasized, in which the ultimate goal of treatment must be clearly defined with realistic expectations.

56.7 TREATMENT MODALITY

56.7.1 Endovascular embolosclectherapy

56.7.1.1 GENERAL OVERVIEW

Endovascular embolosclectherapy with embolization and sclerotherapy modalities have major roles in the “surgically inaccessible” lesion, in addition to a new role to supplement the surgical therapy of a “surgically accessible” lesion.^{1,23}

Endovascular therapy is now the “preferred” therapeutic option in the majority of “extratruncular” AVM lesions, and embolosclectherapy alone is the treatment of choice as an independent therapy for surgically “inaccessible” lesions (Figure 56.1), especially for the “diffuse infiltrating” type with extension beyond the deep fascia, involving muscle, tendon and bone, with prohibitive surgical risks.^{1,2,4,7,16,23,43}

Nevertheless, many new associated problems have been reported, which include both acute complications (e.g., tissue necrosis, vein thrombosis, pulmonary embolism, nerve damage, and cardiopulmonary arrest) and various chronic complications (e.g., muscle/tendon contraction), in addition to their related morbidity.⁴

Therefore, careful assessment of the potential risk of collateral damage is critical in the selection of any embolic agent and sclerosing agent. Every effort should be made to minimize collateral damage to the surrounding tissues—nerve, vessel, cartilage, skin, and soft tissue.

Embolosclectherapy should be performed in multiple sessions whenever possible, using the minimal effective amount of agent during each session in order to minimize the associated risk of complications (e.g., ethanol).

The preferred treatment modality should be selected based on a careful risk–benefit analysis in which the associated morbidity is justified, such as in the case of the treatment of a life-threatening or limb-threatening condition (e.g., hemorrhage or high cardiac output failure).

For primary control of the AVM lesion, absolute ethanol, Onyx, N-butyl cyanoacrylate (nBCA), contour particles, and/or venous coils can be utilized in various combinations,^{42–46} simultaneously or in multiple stages. The use of nBCA or Onyx alone, however, is generally inadequate to provide long-term control of the AVM.^{4,10,44,45}

All of these agents can be grouped into two different categories—embolic agents and liquid agents—based on their chemical properties. All “embolic” agents (e.g., coils) have a limited function in terms of producing mechanical occlusion of the vessel lumen and do not have the ability to penetrate the lesion nidus. In contrast, liquid agents (e.g., ethanol) do have this critical ability to penetrate and treat the lesion nidus. These liquid agents are ideally suited for the treatment of extratruncular AVM lesions.^{4,46,47}

Liquid agents are further classified to two different groups: sclerosants and polymerizing agents. nBCA and Onyx are the two most commonly used polymerizing agents.

Among the many sclerosing agents, ethanol is by far the most commonly used and most effective agent, especially for AVM lesions, while the rest of the sclerosing agents (e.g., sodium tetradecol sulfate, polidocanol, and bleomycin) are primarily used in the treatment of low-flow venous and lymphatic lesions.^{4,23}

Absolute ethanol still remains the sclerosing agent of choice for the treatment of extratruncular AVM lesions, especially in situations in which the lesion is surgically unresectable, such as in the diffuse infiltrating type.^{4,48,49}

56.7.1.2 N-BUTYL CYANOACRYLATE^{4,23,44}

nBCA is a free-flowing adhesive liquid that polymerizes on contact with any ionic solution. Following the immediate mechanical effect to occlude the vessel lumen, nBCA causes an acute inflammatory response to the heat generated during the polymerization process, and then a chronic inflammatory response to its chemical effect.

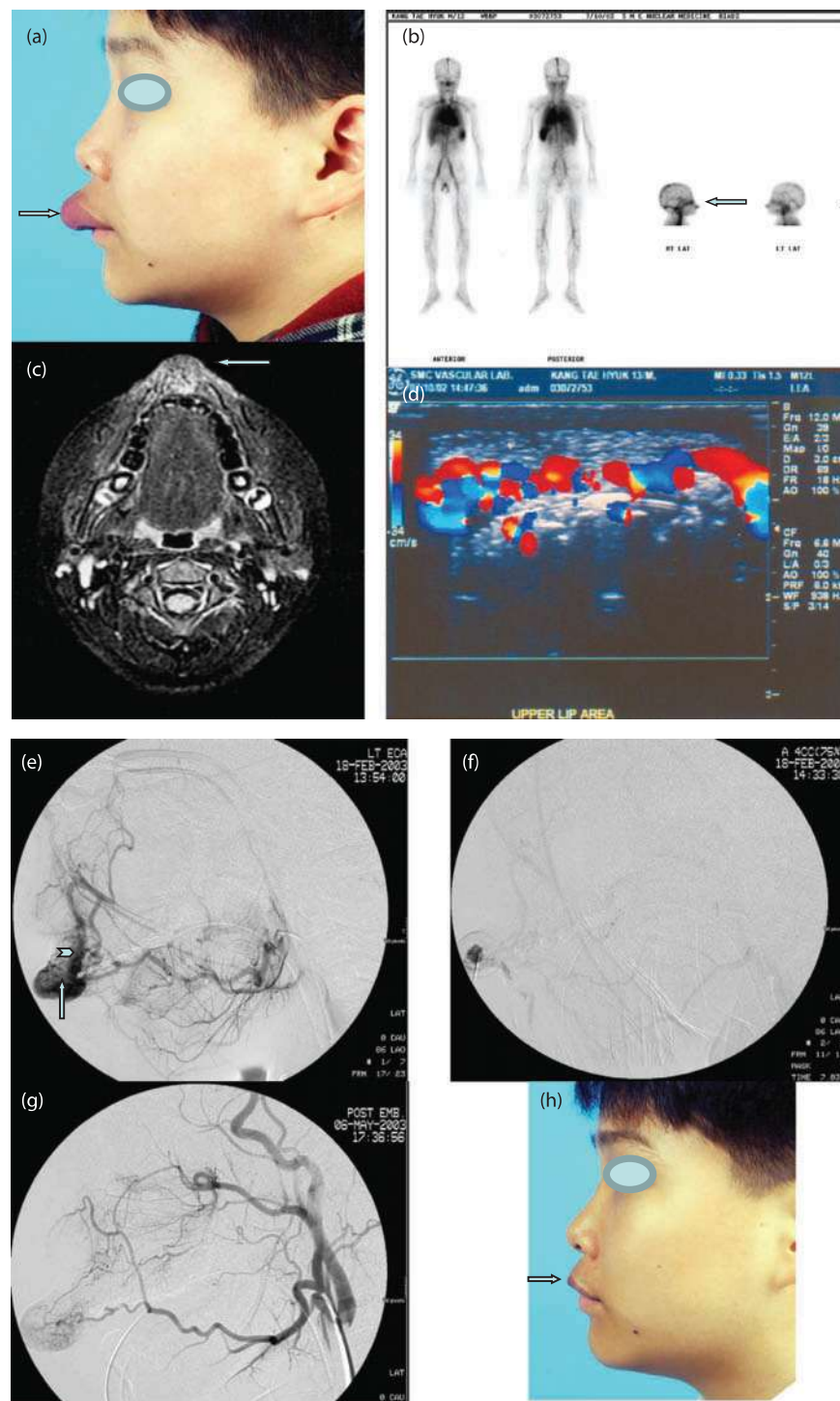


Figure 56.1 Surgically “inaccessible” extratruncular arteriovenous malformation (AVM) lesions treated with multisession emboloscclerotherapies as independent therapies. **(a)** Clinical appearance of AVM lesion affecting the upper lip (arrow) with painful swelling. **(b)** Whole-body blood pool scintigraphy findings of AVM lesion localized to the upper lip (arrow); this test can provide qualitative as well as quantitative measurement for the follow-up assessment. **(c)** T2-weighted magnetic resonance imaging finding of infiltrating extratruncular AVM lesion (arrow) throughout the entire upper lip. **(d)** Duplex scan findings of hemodynamically very active AVM lesion affecting the entire upper lip. **(e)** Pre-treatment arteriographic findings of an upper lip AVM lesion (arrow) with localized nidus and its extensive collaterals as well as its venous drainage (arrow head). **(f)** Angiographic finding of initial ethanol sclerotherapy via the direct puncture percutaneous approach; 4.0 mL of 75% ethanol was used to control the nidus. **(g)** Arteriographic finding of a completely controlled AVM lesion following two sessions of endovascular treatment. **(h)** Clinical appearance of the upper lip with completely restored normal contour following successful ethanol sclerotherapy as an independent treatment.

nBCA has been used as a single-agent therapy with acceptable results in limited specific situations (e.g., inoperable pelvic AVMs). It should not be used as a single-agent therapy in the vast majority of AVM lesions requiring therapy for significant symptoms. Furthermore, nBCA appears to be “resorbed” over time, resulting in AVM recurrence.

Many consider nBCA to be “palliative at best,” and its long-term effects as a foreign body and its effect on the AVM lesion remain controversial when used as a single-agent therapy.

56.7.1.3 ONYX^{4,45}

Onyx is a new, “less adhesive” liquid polymerizing embolic agent consisting of ethylene copolymer and vinyl alcohol dissolved in dimethyl sulfoxide.

Onyx has several advantages over glue (nBCA) for the embolization of AVMs. It is less adhesive and polymerizes more slowly than glue. The microcatheters are rarely glued into the nidus and consequently make intranidal injection more controllable, resulting in a much more satisfactory result. In addition, Onyx carries a much lower risk of pulmonary embolism compared to the other emboloagents.

In some patients, however, onyx embolization results in extensive arterial occlusion without penetration of the nidus of the extratruncular AVM lesions, and so its effectiveness may be variable.

Onyx is not curative and remains a palliative or preoperative embolic agent like nBCA for the treatment of peripheral AVM lesions.

There is a concern regarding neovascular stimulation following extensive Onyx embolization of large AVMs. It is believed that Onyx treatment leads to the development of new collaterals, producing a “massive network of small arteries/arterioles” that is impossible to embolize by any means.

56.7.1.4 COILS^{2,4,7,10,23,50}

Coils are designed to focally occlude larger vessels, rather than penetrating into the lesion nidus. The mechanism of action of coil embolization is limited to the vessels in which it is placed.^{2,4,50}

Therefore, coil embolotherapy itself produces only a mechanical effect to occlude the flow and induce thrombosis. Coils do not have any direct effect on the endothelium and do not prevent recanalization of the lesion nidus. Subsequent regeneration or recovery of the endothelium will result in recurrence of the lesion. Hence, additional permanent therapy is often required to control the nidus completely, either with absolute ethanol or surgical excision if feasible, combined with nBCA.

Coil embolotherapy as an independent therapy is never clinically appropriate for extratruncular AVM lesions. However, coil embolotherapy is the most effective method to convert a high-flow lesion to a lower-flow lesion, which is essential for the treatment of the large, multifistulous AVM lesions that are often associated with a prohibitively high risk of complications related to extremely fast-shunting blood flow and volume.^{2,4,50}

Therefore, a new endovascular approach utilizing initial coil embolization has been developed in which control of the high-flow lesion is achieved with the use of coils, followed by the administration of liquid endovascular agents (e.g., ethanol). Preliminary conversion of a high-flow state to a reduced/lower-flow state would make the lesion more amenable to ethanol sclerotherapy or nBCA glue embolotherapy for subsequent surgical excision with a reduced risk of complication and morbidity.

Hence, coils are typically used as secondary devices on the venous side of the lesion to slow down the outflow in order to enable subsequent injection of absolute ethanol so as to bring about permanent occlusion of the AVM (Figure 56.2).

In contrast, coils are appropriate and very effective for treating “truncular-type fistulous AVM lesions” as the primary treatment. Unlike the extratruncular lesions, the truncular AVM lesions lack the ability to proliferate, so coils can be used as an effective treatment modality, together with other mechanical occlusive devices (e.g., the Amplatzer device).

These truncular lesions often require many coils for successful occlusion of the fistula (lesion) in order to control the associated hemodynamic sequelae (cardiac failure, arterial insufficiency, and venous insufficiency).

56.7.1.5 ABSOLUTE ETHANOL^{46–51}

Absolute ethanol is a powerful sclerosant with curative potential to denude the endothelial cell from the vascular wall, precipitate its protoplasm, and fracture the vascular wall to the level of the internal elastic lamina. The thrombotic process progressively occludes the vascular lumen from the vascular wall, permanently eliminating the “chemotactic cellular factor” and “angiogenesis factor” since the endothelial cell is completely destroyed.⁴

However, ethanol sclerotherapy carries a significant risk of cardiopulmonary complications, warranting appropriate measures to be taken during its administration, including close monitoring with a Swan–Ganz catheter under the general anesthesia.^{49,51}

Pulmonary hypertension is a potentially fatal complication that is caused by pulmonary arterial spasm when a significant amount of ethanol reaches the pulmonary circulation. The pulmonary hypertension can lead acute right heart failure and progress to cardiopulmonary arrest.

Prompt recognition and immediate cessation of ethanol therapy in addition to the appropriate control of pulmonary spasm with the various vasodilators is absolutely necessary to prevent further progression to cardiopulmonary arrest.^{4,49,51} Increased pulmonary artery pressure is the earliest indication of the development of potentially fatal pulmonary hypertension.

A total dose of ethanol of 1 mL/kg of body weight is generally accepted as the maximum volume that can be safely administered during a procedure in order to prevent pulmonary hypertension.

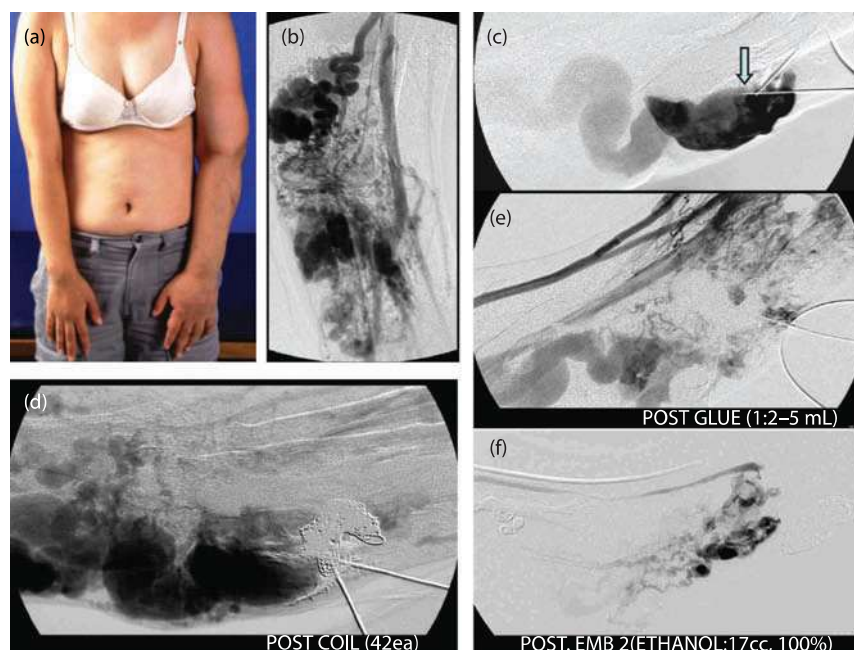


Figure 56.2 Multistage embolosclectherapy of multifistulous arteriovenous malformation (AVM) lesions to control massive high flow. **(a)** Clinical appearance of the left upper extremity affected by AVM lesions along the elbow and forearm region as the cause of massive recurrent bleedings. **(b)** Arteriographic findings of extensive AVM lesions in the “multifistulous high-flow” condition; the feeding arteries are severely dilated and tortuous and the draining veins are also massively dilated. These findings suggest that the lesion has extremely high flow, increasing the risk of the therapy. **(c)** Angiographic finding of a massively dilated vein by a high-flow fistulous connection (arrow) allowing massive shunting from the artery. This condition is extremely difficult to manage with a conventional one-stage approach; it often requires a multistage approach to control/lower the flow with coils and/or glue first before the final treatment with ethanol. **(d)** Angiographic finding of initial coil embolotherapy right at the junction of the arteriovenous connection, making an artificial dam to control the flow. **(e)** Angiographic finding of subsequent N-butyl cyanoacrylate glue embolotherapy to convert the flow to the lowest possible level before final treatment with ethanol. **(f)** Angiographic finding of the final stage of endovascular treatment on the “multifistulous” AVM with 17 mL of 100% ethanol.

Further limiting ethanol administration to 0.14 mL/kg ideal body weight every 10 minutes is recommended when using large volumes of ethanol to treat large lesions in order to obviate the need of a pulmonary artery catheter, as advocated by Shin et al.^{4,49,51}

The safe use of ethanol in AVM sclerotherapy requires precise delivery into the lesion with the use of microcatheters. Proximal injection of ethanol into a feeding artery would cause severe tissue necrosis.

Special precaution should be exercised in order to minimize the increased risk of skin necrosis, especially when ethanol is delivered via an arterial puncture. The minimally effective amount and concentration of ethanol should be administered whenever possible to reduce the risk of complications. Absolute ethanol may be diluted to 60% when used to treat superficial AVM lesions with a high risk of skin necrosis and AVM lesions in close proximity to nerves.

Administration of smaller volumes in divided doses also minimizes the risk of surrounding tissue injury. The residual ethanol within the lesion may be drained before the removal of catheters, especially when swelling/reaction is severe. Light compression may be applied for 5–10 minutes after the removal of catheters following the completion of therapy.

Compression may be applied to the draining vein during the ethanol injection in order to prevent early and premature drainage of ethanol from the lesion and to prolong the exposure time to the endothelium in order to maximize its effect. A great deal of caution should be exercised in order to avoid unnecessary collateral damage.

A combination of the three delivery routes should be utilized to reach the “nidus” of the AVM lesion: transarterial, transvenous, and direct puncture. These approaches should be utilized simultaneously in order to maximize the efficacy of the treatment. The percutaneous direct puncture approach to the lesion, however, is generally preferred and carries minimal risk.

The transarterial approach carries a higher risk of complications; therefore, its use should be limited to situations in which the direct puncture approach is not possible owing to small lesion size (e.g., facial AVMs). In these situations, the transarterial delivery of glue would be a safer approach if indicated and would limit the high risk of skin necrosis.

With the transvenous approach, coil embolization can be performed on the large, high-flow draining vein first, if present, before ethanol is injected.

56.7.2 Surgical/excisional therapy^{2,4,7,10,23,32,43}

Surgical excision has long been the gold standard for the treatment of “extratruncular” AVMs and still remains the most ideal treatment to produce a “cure.” However, unless the lesion is well localized, allowing minimal morbidity with total surgical excision, total surgical removal often carries prohibitively high morbidity and complication rates (e.g., massive operative blood loss and functional loss). Indeed, surgical therapy alone often results in *incomplete* control of the lesion owing to the high morbidity associated with complete radical surgical resection to prevent recurrence.^{2,4,7}

The traditional role of surgical resection has been changed with the development of endovascular therapy over the last decades. A new endovascular embolosclectherapy was introduced as an adjunct to open surgical therapy.^{2,4,7,23,43}

Pre-operative embolosclectherapy results in a significant improvement in the safety and effectiveness of subsequent surgical therapy, producing reduced morbidity and complications associated with surgical resection (e.g.,

intra-operative bleeding). Post-operative supplemental therapy also results in improved overall efficacy of surgical therapy. When implemented for the treatment of residual lesions after surgery, such post-operative embolosclectherapy has also been shown to be as effective as pre-operative embolosclectherapy.

nBCA glue is primarily used pre-operatively in surgically excisable lesions for this purpose. The glue-filled lesion can be safely dissected for surgical removal with minimal collateral damage.^{2,4,10}

Embolosclectherapy is now fully integrated with conventional surgical therapy for the treatment of “surgically accessible” AVM lesions, resulting in improved outcomes compared with lesions treated by surgical therapy alone, especially among marginally accessible lesions (Figure 56.3).

Unfortunately, however, most of the currently available treatments still carry a significant risk of complication and morbidity. Careful planning, from diagnosis and treatment to long-term follow-up assessment, is critical for successful AVM management.^{2,4}

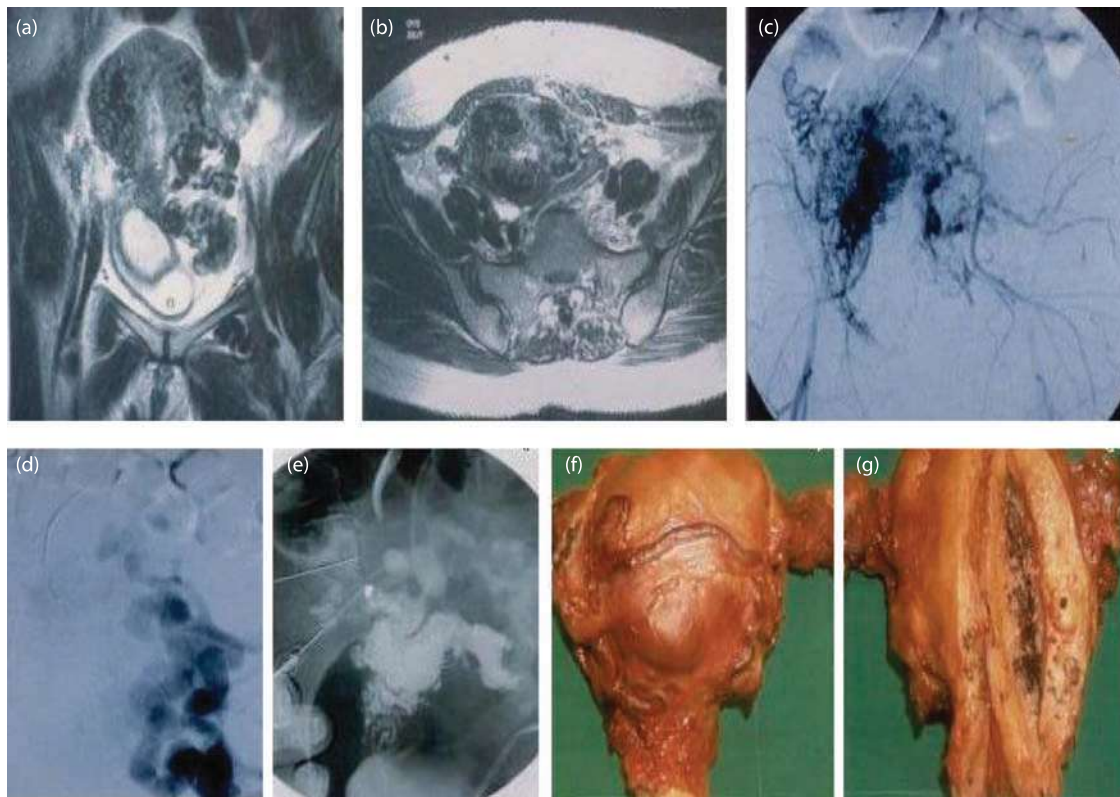


Figure 56.3 Surgically “accessible” extratruncular arteriovenous malformation (AVM) lesion treated with a combined approach with multisession pre-operative endovascular treatment and subsequent surgical excisions. **(a and b)** T2-weighted magnetic resonance imaging (MRI) findings of a pelvic AVM extensively affecting the uterus and para-adnexal soft tissues; this is a life-threatening lesion causing hemorrhagic shock, requiring emergency measures to control massive recurrent uterine bleeding. **(c)** Arteriographic findings of massively infiltrating extratruncular AVM lesions affecting the para-adnexal tissue and uterus. **(d)** Arteriographic finding of massively dilated pelvic veins as the venous drainage route for the extensive pelvic AVM lesions. **(e)** Arteriographic findings of N-butyl cyanoacrylate glue-filled pelvic AVM lesions; this pre-operative embolotherapy reduces/eliminates the risk of massive intra-operative bleeding during the subsequent surgical excision of the lesions. **(f)** Surgical specimen findings of the uterus. **(g)** The inner lumen of the transected uterus also shows glue-filled lesions infiltrating all along the endometrium as well as the uterine muscle structures, which is compatible with the MRI findings.

Periodic follow-up evaluation and assessment of treatment results should be made based on duplex scan, whole-body blood pool scintigraphy, TLPS, computed tomography, and/or MRI in the majority of cases. This is especially important during therapy requiring multiple treatment sessions. For evaluation of the majority of AVMs, arteriography has been the gold standard for the confirmation of treatment results at its completion.^{2,4}

56.8 CONCLUSIONS

A multidisciplinary team approach is required to achieve effective control of AVM lesions with fully integrated

endovascular therapy and surgical therapy. Endovascular therapy is preferred in the surgically “inaccessible” lesion, whereas surgical therapy combined with supplemental endovascular therapy is the best option for the surgically “accessible” lesion.

An early aggressive approach to all AVM lesions is required in general in order to reduce the consequences of any hemodynamic impact. A calculated approach is warranted based on a careful assessment of the risks and benefits associated with the treatment, unless the treatment is indicated for a life-threatening or limb-threatening condition (e.g., hemorrhage or high cardiac output failure).

Guidelines 5.4.0 of the American Venous Forum on arteriovenous malformations: evaluation and treatment

No.	Guideline	Grade of recommendation (1: strong; 2: weak)	Grade of evidence (A: high quality; B: moderate quality; C: low or very low quality)
5.4.1	For symptomatic arteriovenous malformations, we recommend endovascular treatment with embolization or sclerotherapy. We recommend it for both definitive treatment of surgically “inaccessible” lesions and for initial therapy of surgically “accessible” lesions.	1	B

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● = Seminal primary article
★ = Key review paper
◆ = First formal publication of a management guideline

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The management of venous malformations

JOVAN N. MARKOVIC AND CYNTHIA K. SHORTELL

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57.1 INTRODUCTION

Congenital vascular malformations (CVMs) are rare, localized, or diffused abnormalities of vasculogenesis and angiogenesis that arise by embryologic dysmorphogenesis without increased endothelial proliferation, which leads to true structural and/or functional anomalies of the vascular system.^{1,2} Most clinicians—from primary care doctors to subspecialists (including vascular surgeons)—consider the management of CVMs to be a difficult task reserved for referral centers with specialized expertise in this area. The main reason why expertise for vascular malformation management is centralized to major centers is the low frequency at which they occur, the confusing nomenclature and the lack of a uniform classification system that traditionally characterized the majority of the literature discussing CVMs, as well as the absence of established guidelines in the past for their management. Some malformations were initially described according to their appearance (nevus flammeus, port wine stain, stork bite, etc.).³ Other descriptions relied on pathology of the abnormality (angioma simplex, hemangioma cavernosum and recemosum, or glomangioma).⁴ Consequently, a significant number of CVM patients have been discouraged by the lack of correct diagnosis and proper treatment, despite numerous visits to different clinics.

The current chapter will deal with the diagnosis and management of low-flow vascular malformations (LFVMs) that are considered to be predominantly venous. With an estimated incidence of 0.8%–1%, predominantly venous malformations are the most common type and they comprise approximately two-thirds of all CVMs.⁵ The evaluation and treatment of high-flow vascular malformations (HFVMs) with clinically significant arteriovenous shunting is described in [Chapter 56](#).

57.2 CLASSIFICATION

As mentioned earlier, historically, numerous attempts have been made to properly classify CVMs based on anatomic, clinical, and/or embryologic criteria, and no real consensus existed regarding the nomenclature and management of these lesions. Based on a classification of “vascular birthmarks” initially proposed by Mulliken and Glowacki in 1982,¹ the International Society for the Study of Vascular Anomalies (ISSVA) introduced a classification system in which all vascular anomalies were divided into two categories: vascular malformations and vascular tumors. The differentiation of vascular anomalies into tumors and CVMs permitted more effective communication between different medical specialists. Unfortunately, the eponym-based terminology which characterized the ISSVA classification, as well as the lack of clinical applicability of this classification with regards to pre-treatment planning, was believed by many to be a major limitation for widespread acceptance and utilization of this system. To address the ISSVA classification limitations, the authors of “Hamburg” and subsequently the “Modified Hamburg Classification” developed a classification system by incorporating CVM flow characteristics (i.e., arterial, venous, and capillary) and defining CVMs based on the stage of developmental arrest during embryogenesis into truncular and extratruncular lesions, which is important for treatment selection and prognosis of treatment outcomes, as extratruncular lesions are associated with significantly higher recurrence rates and resistance to therapy, presumably because of their preserved mesenchymal characteristics of independent growth potential.⁶ The “Modified Hamburg Classification” system has been considered to be “a more clinician-friendly classification for CVM management,” as outlined in the 2009 Consensus

Document for the treatment of venous malformations of the International Union of Phlebology (IUP).⁷

Despite these efforts, many physicians still did not understand the difference between CVMs and vascular neoplastic lesions, and archaic terms such as “cavernous hemangioma,” “port wine stains,” “angel kiss,” etc., were still frequently used by some specialists to describe CVMs. The growing recognition of the need for more efficient communication among practitioners and the need for a clinically applicable classification system informed the creation of a unified classification system by a group of authors commissioned by the IUP to foster comprehensive management of CVMs. The authors used both the ISSVA and “Modified Hamburg Classification” systems as the basis for the development of a new Integrated Classification System for the management of CVMs.⁸ Malformations were categorized according to hemodynamic characteristics either as high flow or low flow, with further subdivision into anatomic subgroups designated by the predominant vascular element (arterial, venous, capillary, or lymphatic). In this scheme, venous, lymphatic, and capillary malformations are all low-flow lesions. Arterial malformations are considered high-flow lesions. Malformations are further classified either as extratruncular or truncular. Extratruncular lesions are categorized as diffuse/infiltrating or localized, whereas truncular lesions are categorized as obstruction/narrowing or dilatation (Table 57.1). In addition, vascular malformations usually occur alone, but there are also complex combined CVMs that are typically associated with various extravascular anomalies. A detailed description of these syndromes is beyond the scope of this chapter, but the reader is referred to an excellent review by Garzon and colleagues.⁹ The relatively recently introduced concept of the multidisciplinary team approach, characterized by full integration of the expertise of different medical specialists, mandates the use of a unified classification system as an initial step in a multistep algorithm for the management of CVMs.⁸

57.3 ETIOLOGY

Although the etiology of CVMs remains to be elucidated, data from relatively recent studies suggest that the pathophysiologic mechanisms responsible for the formation of CVMs are caused by dysfunctions in the signaling process(es) responsible for the regulation of the proliferation, differentiation, maturation, adhesion, and apoptosis of vascular cells during the development of the vascular system.¹⁰ Vessel development occurs in two different ways: vasculogenesis and angiogenesis.¹¹ The exact mechanisms through which these processes occur remain largely unknown and have just recently begun to be unveiled. Vasculogenesis refers to the process by which endothelial cells are differentiated *de novo* from mesodermal precursors. It occurs only during embryonic development. In angiogenesis, new vessels are formed from pre-existing ones by budding (sprouting), splitting (intussusception), and fusion (intercalated growth). These new vessels that are formed by angiogenesis (the so-called juvenile system) evolve into mature vessels by the processes of maturation and remodeling.¹² These complex processes involve several receptor tyrosine kinases. Some of these receptors and their ligands have been identified (i.e., vascular endothelial growth factors and Tie1 and Tie2).¹³

Several specific genetic abnormalities affecting some of these receptors in families with malformations have been identified. The study of these mutations has not only improved the understanding of the molecular pathogenesis of vascular malformations, but also facilitated the definition of each subtype into more specific clinical entities. For example, genetic analysis of families with autosomal dominantly inherited cutaneous venous malformations revealed mutations on the gene encoding for the kinase domain of the endothelial cell receptor Tie2 (also known as TEK) located at the *VMCM1* locus on 9p2.^{14,15} This receptor has been associated with three ligands: angiopoietins 1, 2, and 4. It is theorized that mutations of this gene produce increased

Table 57.1 Integrated classification system of congenital vascular malformations^a

Congenital vascular malformations					
Low flow			High flow		
Venous	Extratruncular	Diffuse/infiltrating	Arterial	Extratruncular	Diffuse/infiltrating
		Localized			Localized
	Truncular	Obstruction/narrowing		Truncular	Obstruction/narrowing
		Dilatation			Dilatation
Lymphatic	Extratruncular	Diffuse/infiltrating	Arteriovenous	Extratruncular	Diffuse/infiltrating
		Localized			Localized
	Truncular	Obstruction/narrowing		Truncular	Obstruction/narrowing
		Dilatation			Dilatation
Capillary					

Note: Syndrome associated malformations can be both low and/or high flow vascular malformations.

^a International Union of Phlebology and International Union of Angiology endorsed.

phosphorylation of TEK and inhibition of apoptosis, resulting in abnormal remodeling with uncontrolled sprouting and branching of the capillaries and small veins, but not the larger veins and arteries.¹⁶ Glomovenous malformations, when familial, have an autosomal dominant mode of inheritance, with incomplete penetrance. Thirty mutations have been identified in 86 families affecting the glomulin gene on chromosome 1p21–22.¹⁴ Glomulin expression is limited to vascular smooth muscle cells and it is believed that, when lacking, abnormal differentiation towards the “glomus cell” phenotype occurs.¹⁷ Many other genetic abnormalities have been discovered in families affected by CVMs.

Recently, genetic studies have identified high-susceptibility genetic loci associated with a development of truncular venous malformations found within the *HLA* locus on chromosome 6p21.32.¹⁸ In particular, the number of copy number variations due to either deletion or duplication was found to be associated with an increased probability of developing truncular lesions. The region contains 211 known genes. Data from more recent genetic studies showed that regulatory genes of vasculogenesis and angiogenesis (*TIE-2/PDGFB*) play important roles in the development of extratruncular venous malformations.¹⁹ In 2013, Rössler et al. performed an immunohistochemical analysis for cluster of differentiation-31 (CD31), D2–40, GLUT-1, and Ki67 in order to differentiate hemangiomas from LFVMs.²⁰ In addition to this, the authors analyzed the expression levels of β 1, β 2, and β 3 adrenoreceptor mRNAs utilizing quantitative real-time polymerase chain reaction. Data from this study showed that all LFVMs showed CD31-positive immunostaining of endothelial cells and negative GLUT-1 staining. Immunostaining for Ki67 was positive in proliferative hemangioma endothelial cells (confirming their growth potential) and negative in venous malformation, which can serve as basis for the molecular differentiation and diagnosis of these two lesions. More importantly, venous malformations did not demonstrate significant expression levels of all three subtypes of β -adrenoreceptor mRNAs.²⁰ This was the first study to provide the genetic evidence of distinctions between hemangiomas and venous malformations based on β -adrenoreceptor subtype mRNA levels and provided valuable data that can potentially reveal the mechanism by which β -blocker medications are reasonable treatment options for certain types of hemangiomas but *not* for venous malformations.

57.4 CLINICAL PRESENTATION

The clinical presentation and course of CVMs are highly variable (in both their extent and severity), depending upon location, size, organ involved, and morphology of the lesion. They are rarely asymptomatic, although some lesions remain quiescent for many years. Their appearance and growth may be caused by trauma, infection, and/or the effects of hormones (during puberty, pregnancy, and menopause), or lesions may occur spontaneously in the absence of triggering factors. It seems likely that antecedent trauma

Table 57.2 Incidence of physical changes in 82 affected extremities

Change	No.	Percentage
Color changes	57	69.5
Erythema	33	–
Cyanosis	24	–
Venous varices	49	59.7
Edema	46	56.0
Increased length	20	24.3
Deformity	9	11.0
Ulceration	8	9.8
Pulse deficit	3	3.6
Bleeding	3	3.6

Source: Adapted from Szilagyi DE et al. *Arch Surg* 1976;111:423–9.

can disturb a previously stable collateral system and unmask a lesion. Although progesterone receptors are identified in venous malformations, the exact mechanism that causes the appearance and progression of a lesion when hormonal levels change remains to be elucidated.²¹

Patients most often seek medical attention for pain, hemorrhage, cosmesis, palpable mass, limb edema, varicosities, thrombophlebitis, and/or other complications of venous hypertension (Table 57.2). In addition, some of patients have a significant reduction in daily functional capacity and quality of life. Venous malformations are typically bluish, soft and easily compressible, non-pulsatile masses that can enlarge in size when the affected extremity is dependent or after Valsalva maneuver. There is no increase in local skin temperature or thrill when the malformation is palpated, and there is no bruit present on auscultation. In patients with very large venous malformations, the affected limb may be warmer than the uninvolved limb, and patients may complain of increased girth and heaviness from increased venous volume. Young adolescents may develop scoliosis and limb-length discrepancies. The finding of lateral leg varicosities, capillary malformations, unilateral limb osteomuscular hypertrophy, or other venous abnormalities in a young person should alert the examiner to the possibility of Klippel–Trenaunay syndrome (KTS; see below). Pelvic malformations are complex and can produce rectal pain, sexual dysfunction, massive uterine bleeding, and ureteral outlet obstruction with hydronephrosis.

57.5 EVALUATION

The most important initial step in the management of congenital vascular anomalies is to differentiate CVMs from hemangiomas, as the clinical course and long-term consequences are distinctly different (Table 57.3). CVMs are present at birth and exhibit normal endothelial cell structure, function, and turnover.¹ They grow proportionately with the child and do not regress over time. In contrast, hemangiomas are true neoplastic disorders and pathohistologically demonstrate an increased endothelial cellular turnover

Table 57.3 Characteristics of vascular birthmarks

Type	Hemangioma	Malformation
Clinical	Usually nothing seen at birth; 30% present as red	All present at birth; may not be evident
Clinical	Rapid post-natal proliferation and slow involution	Commensurate growth; may expand as a result of trauma, sepsis, or hormonal modulation
Clinical	Female-to-male ratio of 3:1	Female-to-male ratio of 1:1
Cellular	Plump endothelium, increased turnover	Flat endothelium, slow turnover
Cellular	Increased mast cells	Normal mast cell count
Cellular	Multi-laminated basement membrane	Normal thin basement membrane
Cellular	Capillary tubule formation <i>in vitro</i>	Poor endothelial growth <i>in vitro</i>
Hematologic	Primary platelet trapping: thrombocytopenia (Kasabach–Merritt syndrome)	Primary stasis (venous); localized consumptive coagulopathy
Radiologic	Angiographic findings: well-circumscribed, intense lobular–parenchymal staining with equatorial vessels	Angiographic findings: diffuse, no parenchyma
Radiologic	–	Low flow: phleboliths, ectatic channels
Radiologic	–	High flow: enlarged, tortuous arteries with arteriovenous shunting
Skeletal	Infrequent “mass effect” on adjacent bone; skeletal hypertrophy rare	Low flow: distortion, hypertrophy, or hypoplasia
		High flow: destruction, distortion, or hypertrophy

Source: Mulliken JB, Young AE eds. *Vascular Birthmarks: Hemangiomas and Malformations*. Philadelphia, PA: W.B. Saunders Co., 1988, 35, with permission.

rate.² Hemangiomas usually manifest during the first several weeks of life, proliferate rapidly, with disproportionate growth relative to the child, and then slowly involute over a period of years.

The second step in the diagnostic algorithm, which follows initial differentiation between CVMs and hemangiomas (as well as other vascular neoplasms), is hemodynamic assessment of the lesion, which leads to differentiation between LFVMs and HFVMs.²² Plain radiographs, computed tomography (CT), angiography, venography, duplex ultrasonography, and magnetic resonance imaging (MRI) are techniques that were traditionally used for this purpose, as clinical evaluation often underestimated the involvement of deep structures (i.e., muscles, bones, joints, or abdominal viscera) and frequently was not sufficient to differentiate HFVMs from LFVMs. Plain films can demonstrate soft tissue and bony hypertrophy, limb-length discrepancy, and phleboliths. Contrast-enhanced CT can identify the location of the venous malformation, bony involvement, vessel ectasia, and aneurysm formation. The true extent of the lesion into the soft tissue, is underestimated on non-contrast CT as only contrast-enhanced vessels opacify.

Duplex ultrasonography is a portable, noninvasive imaging technique that provides both functional and anatomic data in the evaluation of venous malformations. It is particularly helpful in defining the flow characteristics within the anomaly, thus aiding in differentiating between pure venous, arterial, and mixed malformations. On duplex ultrasonography, venous malformations demonstrate monophasic, biphasic, and no detectable flow in 78%, 6%, and 16% of cases, respectively, while the duplex ultrasonography of

HFVMs is characterized by multidirectional blood flow, rapid arteriovenous shunting, and high-amplitude arterial waveforms with spectral broadening.²³ On grayscale ultrasonography, venous malformations appear as hypoechoic or heterogeneous lesions, with anechoic structures visible in <50% of cases.²³ In some cases, flow in venous malformations is only detectible on ultrasonography with compression and release of the malformation.

Although the above-described ultrasonography imaging is useful for confirming a diagnosis, as it is rapid and easy to perform, it is frequently inadequate for demonstrating the extent of the lesion and its relationship with the surrounding anatomic structures. Thus, MRI became the imaging modality of choice in the evaluation of the CVMs.²⁴ High-flow lesions have a significantly different appearance on MRI than venous malformations. They typically demonstrate low-signal regions (“flow voids”) that can be observed on T1- and T2-weighted images. In contrast to this, venous malformations are visualized as high T2-weighted signals (Figure 57.1). The absence of feeding artery dilatation and draining is also indicative of venous malformations. As numerous particulars can interfere with the above-mentioned observations, distinguishing between HFVMs and LFVMs using MRI can be challenging in some more complex cases. For instance, a blood vessel that courses within an imaging plane can give an intraluminal signal on MRI despite fast flow, which can be a falsely positive finding for a venous malformation.²²

The relatively recently introduced dynamic contrast-enhanced MRI (dceMRI) more accurately assesses the flow within the lesion, since it yields more information with

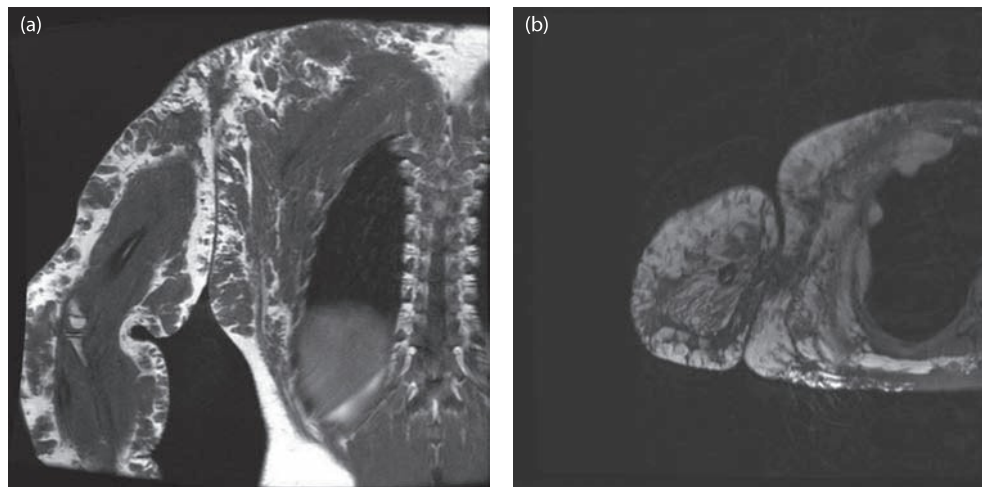


Figure 57.1 Coronal (a) and axial (b) T2-weighted magnetic resonance imaging demonstrates extensive low-flow venous malformation affecting the right hand, arm, shoulder, chest, and upper back in a 9-year-old patient.

regards to the hemodynamic characteristics of CVMs by utilizing time-resolved imaging of contrast kinetics and the time-resolved echo-shared angiographic technique, where images are acquired sequentially every few seconds. These techniques have an additional advantage of being able to delineate dominant or multiple feeding vessels, which can be useful for treatment planning. Data from our study showed a dceMRI accuracy of 83.8% with regards to differentiating HFVMs and LFVMs.²⁵ The data also showed that the specificity and sensitivity for the diagnosis of LFVM were 78.6% and 85.2%, respectively.²⁵ This leaves a relatively small number of inconclusive cases where dceMRI is not definitive for assessing flow characteristics and when suspicion of arterial

flow cannot be excluded based on dceMRI findings. In this subgroup of patients, an appropriate diagnostic workup includes a venography. Venography allows precise evaluation of draining veins and is used to evaluate the extent of a lesion for pre-treatment planning (Figure 57.2). In some patients, angiography can be utilized to provide an opportunity to intervene. Based on data from our analysis, we recommend using dceMRI as the diagnostic modality of choice for making an accurate hemodynamic and anatomic diagnosis and for pre-treatment planning, as dceMRI allows a significant number of patients to be spared the expense, risk, and inconvenience of a catheter-based diagnostic study, as well as a delayed or erroneous diagnosis.²⁵



Figure 57.2 Venography demonstrates a large lateral embryonic vein (arrow) in a patient with Klippel–Trenaunay syndrome. Venography is not only used for diagnosis, but also for the assessment of the patency of the deep venous system and for the communicating pattern between the malformation and the deep venous system, which is important for pre-operative planning. (By permission of the Mayo Foundation for Medical Education and Research. All rights reserved.)



Figure 57.3 Pathognomonic lower extremity hypertrophy, capillary malformations, and lateral varicosities affecting the right lower extremity in a male patient with Klippel-Trenaunay syndrome. (By permission of the Mayo Foundation for Medical Education and Research. All rights reserved.)

57.5.1 Klippel-Trenaunay syndrome

The most common syndrome associated with venous malformations is KTS. This is a rare, sporadic syndrome characterized by the clinical triad of: (1) capillary malformations; (2) soft tissue and bone hypertrophy or, occasionally, hypotrophy of usually one lower limb; and (3) an atypical, mostly lateral varicosity with or without deep venous anomalies (Figure 57.3). It can also include lymphatic anomalies. It is important to differentiate KTS from Parkes-Weber syndrome on the basis that the latter has associated arteriovenous communications and KTS has none. Readers are referred to an exceptional review of the extensive experience at the Mayo Clinic in managing these syndromes published by Gloviczki and Driscoll.²⁶

57.6 TREATMENT

Evaluation with a plan toward intervention in patients with venous malformations is best done by a multidisciplinary vascular malformation team. Prior to treatment planning, every effort should be exerted to rule out the arterial component. This differentiation is of critical value in the management of CVMs, as the treatment options for HFVMs and venous malformations are significantly different and the presence of an arterial component represents an absolute contraindication to sclerotherapy due to the increased risk of

distal embolic events.²² Pre-treatment planning should routinely incorporate the evaluation of the patency of the deep and superficial venous system due to the high prevalence of deep venous system anomalies in venous malformation patients. In a study of 392 patients, Eifert et al. documented aplastic or hypoplastic deep venous trunks in 8% of CVM patients.²⁷ The prevalence of deep venous anomalies is even higher (18%) in the subgroup of patients with KTS.²⁸ This assessment needs to be included in the treatment planning since venous blood flow from the affected extremity may depend on the malformed vessels, and obliteration or exclusion of the malformation from the circulation carries the risk of the impairment of venous return from the affected extremity. Surgical resection of dilated superficial varicosities in a patient with an absent or hypoplastic deep venous system is disastrous. The remaining venous collateral system is not adequate to drain the limb, and massive lower extremity swelling and ulceration can develop.

The treatment of venous malformations is frequently characterized by multiple treatment sessions utilizing the treatment modality that is the most appropriate for the location, extent, and morphology of the lesion. The need for multiple treatment sessions should be discussed with a patient prior to the initiation of treatment to increase the patient's compliance with the therapy. The possibility of recurrence following the treatment (especially in patients with extra-truncular lesions) should also be discussed with the patient prior to starting the treatment in order to minimize the patient's frustration in case of recurrence. Goals of the treatment should be preset with each patient individually, and successful accomplishment of the preset goals marks the completion of treatment, as treatments are frequently palliative and goal oriented, especially in extensive lesions.

If asymptomatic, venous malformation should be treated conservatively (with external compression where appropriate). Patients with venous malformations should be placed in compressive stockings and sleeves early on to offset the long-term complications of venous stasis. Only patients who are symptomatic or have complications of their venous malformation are considered candidates for therapeutic intervention, given the potential for additional morbidity related to any intervention.

Traditionally, surgical resection was effectively used for encapsulated and small lesions. However, in patients with diffuse and multifocal lesions, the surgical approach is relatively contraindicated, as damage to surrounding anatomic structures and massive hemorrhage may ensue. In 2016, Malgor et al. retrospectively evaluated the long-term outcomes of open surgical treatment in 49 patients (53 limbs) with KTS.²⁹ Great saphenous vein stripping and lateral embryonic vein, small saphenous vein, and accessory saphenous vein surgical removal were performed in 17 (32%), 15 (28%), 10 (19%), and nine (17%) lower extremities, respectively. Data from this study showed that two patients developed deep vein thrombosis, one had a pulmonary embolism (PE), and one patient had peroneal nerve palsy. Kaplan-Meier analysis demonstrated that freedom from



Figure 57.4 (a) Large lateral embryonic vein in Klippel-Trenaunay syndrome. (b) Intra-operative photograph demonstrates incisions of a mini-phlebectomy and skin closure with Steristrips following surgical removal of the lateral embryonic vein and congenital varicosities. (From Noel AA et al. Surgical treatment of venous malformations in Klippel-Trenaunay syndrome. *J Vasc Surg* 2000;32:840–7.)

disabling pain at 1, 3, and 5 years was 95%, 77%, and 59%, respectively.²⁹ Respective rates for freedom from secondary procedures were 86%, 78%, and 74%. In addition, at the last follow-up visit, the venous clinical severity score had decreased from 9.4 ± 3.27 to 6.0 ± 3.20 ($P < 0.001$). Data from this study showed that the surgical approach in patients with KTS is safe and durable (Figure 57.4).²⁹

Treatment of the lateral embryonic vein deserves special consideration, since the hemodynamic alterations associated with blood stasis in these frequently valveless veins not only causes symptoms, but is also associated with a high risk of venous thromboembolic events. It has to be emphasized that there is inconsistent terminology that characterizes the majority of the literature discussing persistent embryonic veins. The persistent lateral (marginal) embryonic vein and the persistent sciatic vein are the only two persistent embryonic veins found in KTS patients. The lateral marginal vein is a superficial vein, although from an anatomical standpoint, the term “superficial” is a misnomer, since this vein frequently penetrates the deep fascia and involves muscles of the deep compartment of the lower extremity. The persistent sciatic vein is part of the deep venous system. The specific indications for treatment are based on symptom severity and the status of the deep venous system. In symptomatic patients (i.e., hemorrhage, debilitating pain, and/or functional disability) with patent deep veins, treatment of the lateral embryonic vein is indicated for the elimination of altered

hemodynamics without compromising lower extremity blood return. This also applies to a subgroup of patients with hypoplasia of the deep veins: in this case, treatment of the lateral marginal vein will lead to spontaneous dilation of the deep veins, and therefore hypoplasia of deep veins is not a contraindication for treatment.⁸ By contrast, aplasia of the deep veins does represent a contraindication for intervention, as the lateral embryonic vein serves as a major draining vein of the affected limb in these cases. All patients with a lateral marginal vein should undergo a complete hypercoagulability workup (including measurement of D-dimer and fibrinogen) and should be anticoagulated appropriately with a weight-adjusted dose (100 U/kg/day) of low-molecular-weight heparin peri-procedurally.⁸ Traditionally, as mentioned above, surgical resection is preferred to endovascular treatment modalities as it is safe and durable and due to the superficial location of this vein and the fact that endovenous treatment modalities are associated with a risk of skin damage. In addition to this, in a small series of patients, Frasier et al. documented that the majority of KTS patients treated with radiofrequency ablation required retreatment due to recanalization of the treated vein.³⁰

Definitive cure is possible in patients with limited, superficial lesions that are amenable to complete surgical excision. In reviewing the literature, only 20%–30% of patients with vascular malformations are candidates for complete excision. The largest surgical series were published in 1976, 1983, 1990, and 1992, comprising fewer than 100 patients. Szilagyi et al. reported on 18 patients with vascular malformations treated surgically.³¹ A total of 55% of patients reported improvement in their symptoms following excision, 11% were unchanged, and fully a third were worse after surgery than before.³⁰ In 1992, Scott et al. reported the Northwestern experience in 15 patients with vascular malformations treated surgically.³² Five patients were lost to follow-up. Assuming those patients did well and did not seek further intervention, two-thirds of those undergoing excision improved. A total of 13% were unchanged and 20% were worse after surgical excision.³¹ If complete excision of the malformation is not possible, some authors suggest isolation of the lesion as an alternative. Skeletonization of lesions by ligation of all vessels feeding and draining the lesion is conceptually appealing, but more difficult in practice. Even minor collateral inflow can compensate over time and result in recurrence. Proximal ligation or embolization of major feeding vessels results in temporary improvements of symptoms. Recurrence of the lesion occurs as collaterals restore inflow to the vascular mass. Occlusion of the main feeding trunk surgically or through endovascular means precludes further interventional approaches and should be avoided.

Sclerotherapy has been used as an effective alternative to surgery in the treatment of venous malformations. Traditionally, the most widely used sclerosant was ethanol. Ethanol sclerotherapy, although effective, is associated with limitations and major complications (local and systemic), including severe pain requiring general anesthesia, ethanol toxicity, and localized tissue necrosis.³³ In a study of 87

patients with venous malformations who received 98 sessions of ethanol sclerotherapy, Lee et al. reported outstanding results, with 95% initial success and no recurrence in 71 patients at a mean follow-up of 24 months.³⁴ Data from this study also documented complications in 26.7% of patients which ranged from mild to severe. There were nine cases with ischemic bullae, five with nerve palsy, four cases of transient pulmonary pressure elevation, two with tissue necrosis and tissue fibrosis, one with deep vein thrombosis, and one with PE. Of the five nerve palsies, one (affecting the peroneal nerve) was irreversible.³³ Other studies also reported major side effects following ethanol sclerotherapy, including cardiac arrest and episodes of transient bradycardia.³⁵ Ethanol sclerotherapy can also result in transmural vessel necrosis, massive swelling (sometimes resulting in compartment syndrome), central nervous system depression, hypertension, and pulmonary vasospasm.³⁶ Burrows and Mason reported good to excellent results after serial sclerotherapy in 75%–90% of patients with LFVMs and showed that ethanol injections should be avoided close to major nerves or cutaneous lesions.³⁷ There was a 12% complication rate per session and 28% complication rate per patient in this series, with at least some skin necrosis occurring in 10%–15%.³⁶

Since liquid sclerosing agents become diluted by intralésional blood, the use of sclerosants in microfoam form significantly improves the procedure for venous malformations, as the foam bubbles displace intralésional blood and prevent the sclerosant from becoming diluted. Following intravascular foam injection (with or without ultrasound or fluoroscopic guidance) sclerosants induce irreversible chemical damage to the vascular endothelial lining by the disruption of cell membranes. The response to the subendothelial collagen exposure leads to vasospasm, platelet aggregation, and subsequent endofibrosis that obliterates the vessel.³⁸ Foam bubbles achieve maximal exposure between the sclerosing agent and the endothelial lining of the malformation and the echogenicity of the bubbles makes them visible on ultrasonography, making the procedure more effective and easier to perform. For these reasons, foam treatments, in contrast to liquid sclerotherapy, can be performed on an ambulatory basis under local anesthesia. In a prospective randomized clinical trial, Yamaki et al. compared the efficacy of ultrasound-guided foam sclerotherapy with ultrasound-guided liquid sclerotherapy in the treatment of 89 symptomatic venous malformations.³⁹ Data from this study showed that the amount of sclerosant required to treat the lesion was significantly smaller in patients who were treated with foam sclerotherapy. These properties of sclerosant in the form of microfoam make it possible to use smaller doses in small increments, which reduces the risk of side effects and toxicity.

Foam can be produced with different techniques that result in differences in bubble size, foam stability, and reabsorption rates. The most widely accepted method for producing stable foam is the “Tessari method,” first described in 2001.⁴⁰ The choice of foamed sclerosant is typically physician dependent. Currently, the majority of venous malformations

are treated with 0.5%–1% sodium tetradecyl sulfate or 1% polidocanol. Alternative sclerosants such as bleomycin or doxycycline may be used in special circumstances, including venous malformations located on the hands and in lesions with a lymphatic component. At our institution, we currently use ultrasound- or fluoroscopically guided polidocanol foam sclerotherapy as the first line of treatment for venous malformations in the majority of patients.

The first large study of patients treated by foam sclerotherapy was published by Cabrera et al.⁴¹ This study included 50 patients (35 with venous malformations and 15 with KTS). Sclerotherapy was performed by ultrasound-guided injection of 0.25%–4% polidocanol microfoam. Data from this study showed that the treatment was beneficial in 46 (92%) patients. Total disappearance of treated malformation, reduction in size of more than 50%, and reduction in size of less than 50% were documented in 158, 15, and 13 patients, respectively. In a subgroup of patient who presented with pain ($n = 39$), 25 experienced total relief, and the pain was significantly reduced in the remaining 14 patients. There were no major adverse events. Skin necrosis developed in three patients and four patients developed skin hyperpigmentation that was transient. Data from another perspective study that included 14 LFVM patients (eight with KTS) treated with 1% or 2% polidocanol demonstrated that the use of polidocanol foam sclerotherapy was effective and showed no major complications, no downtime, and no need for general or regional anesthesia.⁴²

Relatively recently, we evaluated the efficacy and safety of the above-described treatment algorithm implemented by our multidisciplinary CVM team in a large series ($n = 136$) of CVM patients.⁴³ There were 105 (77.2%) LFVMs. In total, 47.1% and 14.0% of CVMs were located in the lower and upper extremities, respectively; this was followed by the head and neck (19.9%), trunk (5.1%), and pelvis (2.9%). A total of 11% of CVMs were diffuse, affecting two or more of the above-mentioned anatomic locations. Twenty-three (21.9%) LFVMs were managed conservatively, 38 (36.2%) were treated with sclerotherapy (sodium tetradecyl sulfate, polidocanol, doxycycline, and/or ethanol), 18 (17.1%) were surgically resected, and eight (7.6%) were managed with a combination of modalities. Foam sclerotherapy (sodium tetradecyl sulfate or polidocanol) was used in 31 (29.5%) lesions and ethanol in seven (6.7%) lesions. One (0.95%) patient was treated with doxycycline. Our data showed that conservative treatment was indicated for patients with minimal and tolerable symptoms and uncomplicated lesions. We documented significant improvements in symptoms and the attainment of patient and physician predetermined goals of therapy in approximately 88% of LFVM patients. In a subsequent study, we also analyzed the morphology of LFVMs to identify the lesion characteristics that affect the outcomes of treatment.⁴⁴ Data from our analysis demonstrated that symptomatic, diffuse, extensive LFVMs and LFVMs that involve multiple tissue planes and vital structures are best treated with foam sclerotherapy. LFVMs that are characterized by microcystic, septated vessels do not

respond to sclerotherapy, and these lesions are best treated with surgical resection. Primary surgical resection is also the treatment of choice for localized, microcystic, and superficial LFVMs.⁴³ As mentioned earlier, treatment of extensive lesions was palliative and goal oriented.

Similarly to our findings, based on data from a retrospective review of 33 patients with venous malformations, the group from Birmingham Children's Hospital proposed conservative management for patients with tolerable symptoms, reserving primary resection for localized and well-circumscribed lesions and sclerotherapy for vascular malformations involving vital structures.⁴⁵ Data from this study showed that 3% sodium tetradecyl sulfate sclerotherapy (performed in 12 patients) was associated with a 75% benefit and improvement in symptoms. The three patients who did not benefit from sclerotherapy had malformations infiltrating the knee joint. The cumulative complication rate for sclerotherapy and surgery was 15%. Of those who had sclerotherapy, three patients (25%) developed terminal necrosis of the distal phalanx, skin necrosis, and acute renal failure (due to the administration of a large dose of sclerosant) treated by dialysis. In a subgroup of patients who had surgery, minor complications such as wound dehiscence and skin necrosis occurred in two patients (11%).⁴⁴

The initial determination of whether sclerotherapy can be performed safely must be made on clinical grounds for each patient and each lesion by an experienced practitioner with the knowledge of the venous malformation extent, communicating pattern, hemodynamics, and the feasibility of other treatment modalities. In properly selected patients, sclerotherapy is technically feasible and the risk of severe side effects can be minimized by utilizing several techniques that limit venous outflow during the delivery of the sclerosant. A tourniquet or manual compression can be applied downstream of the venous outflow to minimize the risk of passage of the sclerosant into the systemic circulation, to help contain sclerosant within the lesion, and to ensure maximal contact between the sclerosing agent and the endothelial lining of the malformation.

In the most challenging cases, in which the anatomy of the venous drainage is complex, multimodal (selective) treatment of the venous malformations, including pre-operative embolization of the draining vein followed by sclerotherapy of the venous malformation, can be effectively

utilized. In a study of 201 patients, Jin et al. showed that embolization of the draining vein followed by sclerotherapy of the lesion was beneficial in 196 (97.5%) patients.⁴⁶ Among the responders, complete resolution of symptoms, significant improvement (to nearly normal), and marked improvement were documented in 56 (28.6%), 42 (21.4%), and 62 (31.6%) patients, respectively. After a total of 592 treatment sessions, the authors documented no major complications (defined as side effects requiring further therapy, permanent adverse sequelae, or death). Minor complications were sclerosant specific and included tissue necrosis, peripheral nerve palsy, blistering, skin pigmentation, fever, and gastrointestinal irritation, which were documented in 6 (1.0%), 5 (0.8%), 5 (0.8%), 10 (1.7%), 35 (5.9%), and 76 (12.8%) cases, respectively. All complications were self-limiting and were characterized by spontaneous resolution.⁴⁵

57.7 SUMMARY

Over the last two decades, there has been significant progress in the understanding of the molecular and genetic mechanisms responsible for the etiology and pathophysiology of venous malformations and of the management of these lesions. Data from numerous studies provide strong evidence that experienced providers working in the context of a coordinated and structured multidisciplinary vascular malformation team offer safe and efficient management of patients with CVMs. The diagnostic algorithm that is utilized to distinguish vascular tumors from CVMs and venous malformations from HFVMs is clinically applicable for making an accurate diagnosis and for pre-treatment planning in the majority of patients. A systematic evaluation of venous malformation is based on the characteristics of the blood flow, delineates the anatomic extent of involvement, and determines the presence of associated extravascular anomalies. For most asymptomatic venous malformations, observation alone is the best strategy. For others, external compression alone is sufficient treatment. Patients with symptomatic venous malformations are best treated with foam sclerotherapy and/or primary surgical resection, dependent upon the lesion's location, extent, and morphologic characteristics. This treatment pathway results in favorable outcomes with reasonable complication rates in this frequently challenging group of patients.

Guidelines 5.5.0 of the American Venous Forum on the management of venous malformations

No.	Guideline	Grade of recommendation (1: strong; 2: weak)	Grade of evidence (A: high quality; B: moderate quality; C: low or very low quality)
5.5.1	For symptomatic venous malformations not responding to compression treatment, we suggest foam sclerotherapy over sclerotherapy with alcohol.	2	C
5.5.2	For surgically accessible and localized symptomatic venous malformations, we suggest surgical excision as an alternative to sclerotherapy.	2	C

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